CS496 AI Engineering
Project 0: The Bake-Off

Legal Clause Classification

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1. Task

The task is **legal clause classification**. Given a short contract clause, the model must classify it into exactly one of ten predefined categories:

- CONFIDENTIALITY
- TERMINATION
- PAYMENT
- LIABILITY
- INDEMNIFICATION
- INTELLECTUAL_PROPERTY
- GOVERNING_LAW
- DATA_PROTECTION
- WARRANTY
- NON_COMPETE

For example, the clause *"The customer shall pay all invoices within thirty days of receipt"* should be classified as `PAYMENT`.

The benchmark uses automatic exact-match scoring. A response is correct only when it contains exactly one valid category. Explanations, invalid labels, multiple labels, refusals, and errors are counted as incorrect.

2. Data and Evaluation

The dataset contains **100 clauses**:

- 20 development items
- 80 test items
- 10 categories
- 10 items per category

Only the 80 test items were used for the final benchmark. Category definitions and labeling rules are documented in `data/labeling_note.md`.

The benchmark uses the same 80 test items for every model.

3. Setup and Models

All three models were run locally using **Ollama**.

Model	Runtime	Hardware
`qwen2.5:7b`	Ollama	NVIDIA RTX 3060 12 GB
`qwen2.5:3b`	Ollama	NVIDIA RTX 3060 12 GB
`phi3.5:latest`	Ollama	NVIDIA RTX 3060 12 GB

The same prompt was used for all models. The generation settings were:

- Temperature: 0
- Maximum generated tokens: 20
- Sequential evaluation
- One model call per item
- Same parser and automatic scoring function

The prompt is implemented in `src/prompt.py`, while automatic scoring is implemented in `src/score.py`.

4. Results

Each model classified the same 80 test items, resulting in 240 total model evaluations.

Model	Correct	Accuracy	p50 (ms)	p95 (ms)
`qwen2.5:7b`	78/80	97.50%	3262	3701
`qwen2.5:3b`	68/80	85.00%	2321	2382
`phi3.5:latest`	77/80	96.25%	2729	3029

Model	Tokens/sec
`qwen2.5:7b`	9.75
`qwen2.5:3b`	39.38
`phi3.5:latest`	19.42

The results show a measurable trade-off between classification accuracy and generation speed. `qwen2.5:7b` had the highest measured accuracy in this experiment, while `qwen2.5:3b` had the highest measured token generation rate. `phi3.5:latest` produced accuracy close to the 7B model while having lower latency than the 7B model.

5. Representative Errors

5.1 `qwen2.5:7b`

Item 044

Expected: `INDEMNIFICATION`

Output: `LIABILITY`

> The service provider shall defend and hold the customer harmless from claims alleging that the services caused bodily injury due to the provider's negligence.

Item 095

Expected: `NON_COMPETE`

Output: `TERMINATION`

> For six months following termination, the executive shall not provide competing services to customers within the defined market.

5.2 `qwen2.5:3b`

Item 038

Expected: `LIABILITY`

Output: `PAYMENT`

> A party's aggregate liability under this agreement shall not exceed fifty thousand euros.

Item 039

Expected: `LIABILITY`

Output: `TERMINATION`

> The service provider shall not be liable for delays caused exclusively by events outside its reasonable control.

Item 044

Expected: `INDEMNIFICATION`

Output: `LIABILITY`

> The service provider shall defend and hold the customer harmless from claims alleging that the services caused bodily injury due to the provider's negligence.

5.3 `phi3.5:latest`

Item 044

Expected: `INDEMNIFICATION`

Output: `WARRANTY`

> The service provider shall defend and hold the customer harmless from claims alleging that the services caused bodily injury due to the provider's negligence.

Item 086

Expected: `WARRANTY`

Output: `INTELLECTUAL_PROPERTY`

> The supplier represents that it possesses all licenses necessary to provide the contracted services.

Item 089

Expected: `WARRANTY`

Output: The response was `You are a legal clause classification system with advanced`.

This response was counted as a parse error because it did not contain exactly one valid category.

6. Model Selection Considerations

The measurements provide different trade-offs rather than a single universal result.

`qwen2.5:7b` recorded the highest accuracy at 97.50%, with 78 correct classifications out of 80. Its p50 latency was 3262 ms and its average generation speed was 9.75 tokens/sec.

`qwen2.5:3b` recorded 85.00% accuracy and the highest generation speed at 39.38 tokens/sec. Its p50 latency was also the lowest at 2321 ms.

`phi3.5:latest` recorded 96.25% accuracy, with a p50 latency of 2729 ms and an average generation speed of 19.42 tokens/sec.

Therefore, the appropriate model depends on the application's requirements. A workload that prioritizes measured classification accuracy may place greater importance on the accuracy results, while a latency-sensitive workload may place greater importance on response time and generation speed.

7. Cost and Scope

This implementation uses three locally hosted Ollama models and does not use external API services or API tokens. Therefore, API token pricing is not included in this comparison. Hardware and electricity costs are also not estimated because this project focuses on the measured local-model performance.

8. Conclusion

The bake-off successfully established a reproducible evaluation pipeline for comparing three locally hosted LLMs on legal clause classification.

Using the same dataset, prompt, generation settings, parser, and scoring procedure produced measurable differences in accuracy and runtime performance. The results demonstrate why both task accuracy and system performance should be considered when evaluating local LLMs.